Functions & Proportionality

Lesson 3 / Overview

Students explore exponential functions after one last comparison between linear and quadratic functions. They compare this new function to linear and quadratic ones. Comparisons are made among rules, patterns in tables, and graphs.

Materials
POD: Overhead POD 3
Student Pages: Exp. Growth of Quad; Why $3^0=1$; Let's Start a Rumor; Paper Folding; Tower Puzzle.
Other Materials:
Graphing calculators, Tower puzzles for each group;

Teaching Actions

POD: *In a 10 by 10 grid, the outside squares are shaded. How many are shaded? If a "m by m" grid had the outside squares shaded, how many squares would be shaded?*

Encourage students to solve simpler problems, build a table and look for patterns. The constant difference in the table shows the function to be linear. [y = 4m - 4].

Review: Go through students' comments from previous day's Big Idea forms. Emphasize differences among tables, rules and graphs for linear and quadratic functions.

Large Group Introduction

1. The quadratic examples shown so far show more rapid growth than for linear functions. Often students over generalize that quadratic growth is more rapid than linear growth. To counter those generalizations, ask students to complete, *Exploring Growth of Quadratic Functions.* In these examples students come to see that the coeficent of x or x^2 affects how fast the functions grow. The critical difference in growth between linear and quadratic is that linear functions grow at a constant rate; quadratic does not. A constant growth is found by looking at the second order differences.

2. End this part of the lesson by asking students to build a table for: $y = 2x^2 + 3x - 4$. Look at the growth in the table. Explain that there are many different forms of quadratic functions, and that they have been examining simple ones.

Comments

1. Different counting strategies will yield different forms of function for today's POD.

Example:

Subtract two from the length; multiply by 4 add 4 = outside squares; [(m-2) x 4 + 4].

Multiply length by 4; this adds the corners twice so subtract 4 to find # of outside squares; [4m - 4].

Outside squares = total number of squares minus all the squares which make up the unshaded square.
$[m^2 - (m-2)^2] = 4m - 4]$.

2. The rest of the lesson introduces the exponential rule. Differences among the three functions are discussed. What is of particular interest is the lack of a constant difference in tables of exponential functions, and the presence of multiplicative relationship going down the table for the Y values

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Teaching Actions

3. Pose the Rumor problem. Be sure students understand how the rumor is being spread. Consider the day when Tom is the only one to know the rumor as Day 0 (so 0,1 is the first data point). Then on day 1, he tells three people, so there are 3 new people who hear the rumor on day 1. On day 2, the new people from day 1 tell 3 new people, so 9 new people know the rumor on day 2. Each day only the new people from the previous day spread the rumor. On day 3 there are 27 new people who hear the rumor.

Small Group Work

4. Have students explore the problem in groups, completing the questions that go along with the table. Strongly encourage students to look for patterns and to talk about them "in their own words".

5. The table students complete will look like the following:

Day	# of new people to hear the rumor on a given day
0	1
1	3
2	9
3	27
4	81
5	243

Large Group Discussion

6. Have students share patterns they found in the table. Discuss how the patterns in this table differ from tables of linear and quadratic functions. [See notes at the right].

7. Point out how the dependent variable increases rapidly. This is characteristic of one type of exponential function.

Comments

3. Students should write down the patterns they talk about. Most will see that each day equals the previous day times 3. If they write: 3x3; 3x3x3; 3x3x3x3 they will be more successful finding the rule.

4. Notice how quickly the dependent variable increases. If the rule was: $y = 1/3^n$ then the dependent variable would decrease rapidly.

5. Constant differences do not occur for exponential functions.

n	3^n				
0	1				
		2			
1	3		4		
		6		8	
2	9		12		
		18		24	
3	27		36		
		54		72	
4	81		108		
		162			
5	243				

There are no constant differences. What is special is the constant multiplicative relationship: 1x3 = 3; 3x3=9; 9x3 = 27; 27x3=81. Also notice the multiplicative pattern in each set of differences.

6. Look at the graph for $y = 3^n$. Note that the line doesn't cross the x-axis. Use the calculator to access the table for the function and scan the table for x = -1, -2, -3...

Discuss that: $3^{-1} = 1/3$; $3^{-2} = 1/9$; 3^n approaches zero as n decreases.

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Teaching Actions

8. Share the rule found for the data [$y = 3^n$ where n = the day]. Name this function as *exponential*. Point out the dependent variable is the exponent. Compare this rule to an example of a linear and quadratic function from the previous 2 lessons.

Note that $y = 3^n$ does equal 1 if n = 0. The idea that any value raised to the zero power is one is not easily accepted and is a source of error when evaluating exponential functions for n = 0.

9. Graph the function using the calculator. Compare the graph to linear and quadratic graphs.

Extension

10. Use the graphing calculator to build tables for other exponential functions. Examine the tables for multiplicative patterns. Then graph the functions.

$y = 2^x + 2$ $y = 3^x - 1$

11. Look for multiplicative patterns in the Y values.

Break

Small Group Work

12. Have students work on **Paper Folding Patterns**. Explain that each group will be called on to explain how they approached the problem.

Large Group Discussion

13. Share students' responses to the paper folding problem. The rules for this problem are:

$y = 2^n$ **$y = 1/2^n$**

Comments

7. $y = 2^x + 2$

x	y		
-1	2.5		
		.5	
0	3		.5
		1	
1	4		1
		2	
2	6		2
		4	
3	10		4
		8	
4	18		

After the first difference, the pattern going down the table shows the numbers increasing by a factor of 2.

$y = 3^x - 1$

x	y		
-1	-2/3		
		2/3	
0	0		4/3
		2	
1	2		4
		6	
2	8		12
		18	
3	26		36
		54	
4	80		

After the first difference, the pattern going down the table shows the numbers increasing by a factor of 3.

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Teaching Actions

14. The important points to discuss are how the patterns in the tables are different from patterns in the tables for linear and quadratic functions. With exponential functions a constant difference is not found as with the other two types. Note how quickly the dependent variable increases or decreases with exponential functions- this is related to the constant multiplicative growth.

Small Group Work

15. Explain the rules for the **Tower Puzzle**. In groups students work through this puzzle. This is also an exponential function. **[$y = 2^n - 1$]**

Large Group Discussion

16. To summarize differences among the 3 functions ask students to build a table and graph each of these functions: $y = -(x+1)$ $y = -2x^2 + 1$ $y = -3^x$

Examine rules, tables and graphs. Ask: What are the characteristics of each type of function.

17. Complete Big Idea form in groups.

Comments

8. Students may need help with evaluating expressions with negat signs.

$y = -2x^2 + 1$

If x = 6 then six is squared first, th multiplied by -2.

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Exploring the Growth of Quadratic Functions

The quadratic functions you have looked at have shown more rapid growth than the linear functions you have examined. Examine the two examples below. How does the dependent variable (y) grow in each one as the independent variable (x) increases by one?

$y = 2x$

X	Y
0	0
1	2
2	4
3	6
4	8
5	10
6	12

+2

$y = 2x^2$

X	Y
0	0
1	2
2	8
3	18
4	32
5	50
6	72

+2

In which case is the dependent variable growing faster?

Examine the table for the quadratic function: $y = .10\ x^2$. Use the graphing calculator to generate the data table:

X	0	1	2	3	4	5	6
Y							

How do the y values grow as the x values increase by one? Does the growth match the pattern of growth in other quadratic functions you have examined? Explain.

Is the growth very fast? Why or why not?

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Build a table of values for $y = .10\ x$

X	0	1	2	3	4	5
Y						

How does the dependent variable, y, increase as the independent variable increases by one? Is this growth faster or slower than the growth of the y values for $y = .10\ x^2$? Explain.

Build a table of points for the $y = 10\ x$

X	0	1	2	3	4	5
Y						

How does the growth of the dependent variable compare to the growth in the table for $y = .10\ x^2$? Explain why.

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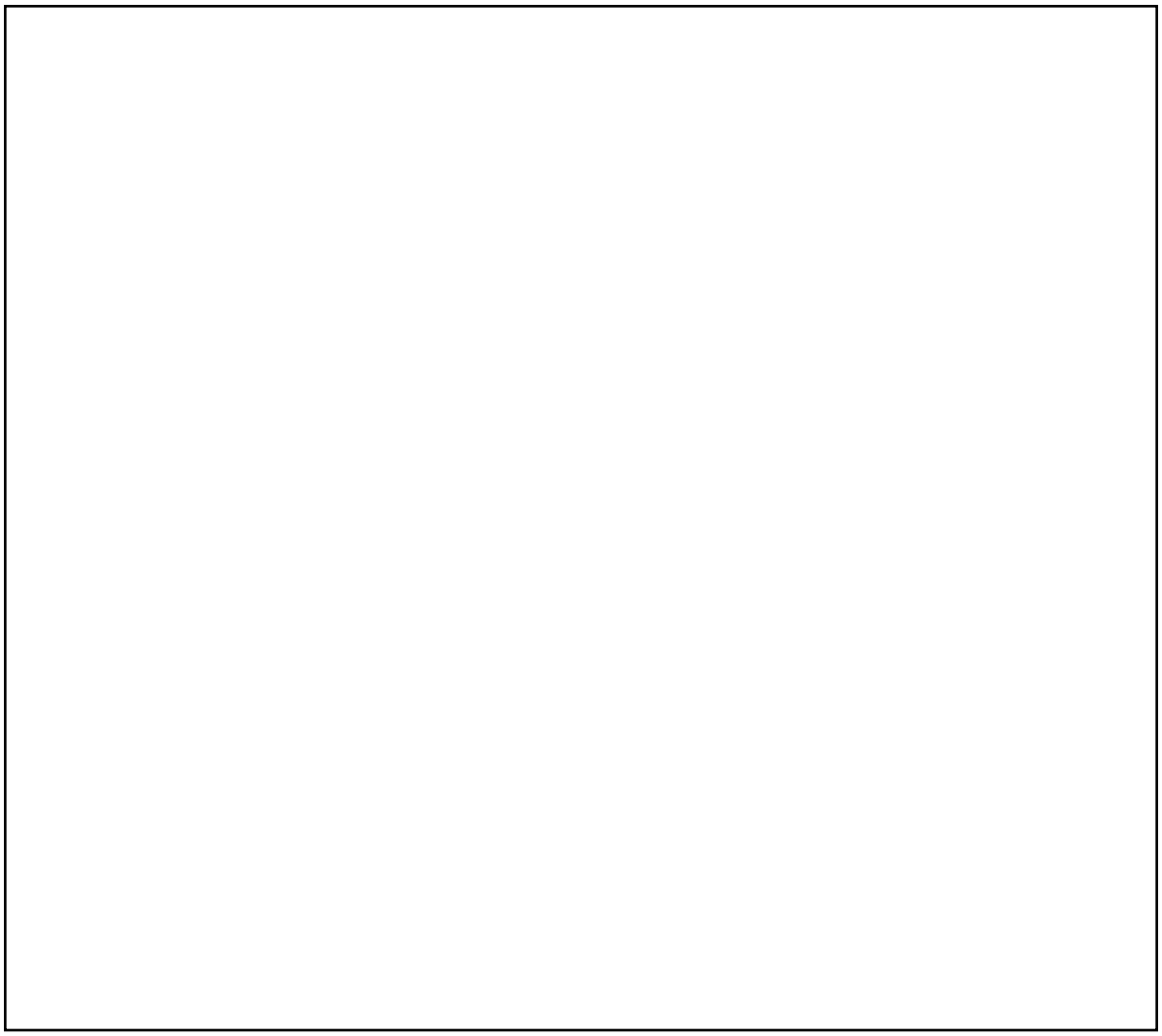
Questions????

- How does the coefficient of x^2 affect the growth of the dependent variable?

- What generalizations can you make about the growth of quadratic functions? What factors affect their growth?

- Check out your generalizations with these functions: $y = .5\ x^2$; $y = .5\ x$; $y = 5x$; $y = 5x^2$.

- Compare the graphs.



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Let's Start a Rumor

In Bugsville, USA, at Swat Middle School, Tom, a second grade teacher decided to start a rumor that the town is going to declare September 14 as National Bug Day and is going to close all schools for the day. He tells 3 people the rumor with instructions that each of these people is to repeat the rumor to 3 more people the next day and that each of these new people is to repeat the rumor to 3 more people on the third day and so on.

Build a table to explore the number of new people who hear the rumor
on a given day.

Day	# of new people who hear the rumor on a given day
0	
1	
2	
3	
4	
5	
6	

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QUESTIONS ??????????

1. Describe patterns that you observe in the table.

2. On what day will 19,683 new people be told the rumor? How many new people will have heard the rumor on day 10? If there are about 65,000 people in the town, by what day will all have heard the rumor?

3. Describe the relationship between the first two columns in the chart algebraically.

4. Graph the relationship. Describe in detail what the graph looks like.

5. Let's step outside of the context. What will the value of y be for x = -1, -2, -3, -4 ? Use your calculator to evaluate for y: [3^(-1) enter]

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Extension:

Use your graphing calculator to build a table for these exponential functions. Record data below each function:

$y = 2^x + 2$ $y = 3^x - 1$

Look closely at the tables. Are there constant differences? Any multiplicative patterns?

Graph each of the functions. Set up the window in the following way:

Xmin = -5
Xmax = 10
Xscl = 1
Ymin = -10
Ymax = 150
Yscl = 2

Describe what the graphs look like:

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Paper Folding Patterns

Take a piece of paper and fold in half as many times as you can. After 1 fold there will be 2 regions. How many regions will occur after 3 folds? four folds? How many folds are possible? Complete the table below. Imagine that there is no limit on the number of folds possible

# of folds	# of regions	
0		
1		
2		
3		
4		
5		

Questions ???????

1. Describe patterns found in the table. Have you seen this table before?

2. Algebraically, describe how the number of regions is related to the number of folds? Use this rule to determine the number of regions given 18 folds.

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3. Add another heading to the table: Area of the smallest region. Complete the table for this heading.

4. Describe patterns observed in the table. How is the area related to the number of folds? Describe this relationship algebraically.

5. Graph both relationships. How are they alike? Different?

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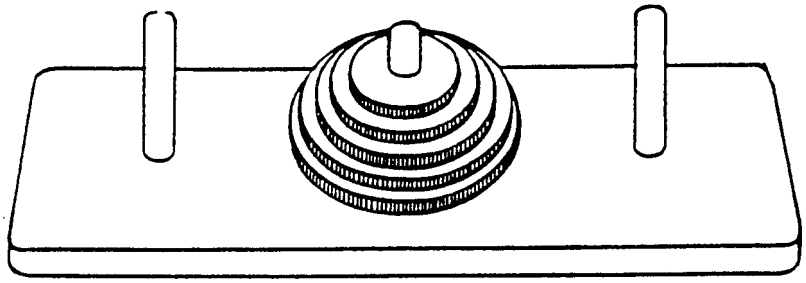
The Tower Puzzle

The object of this puzzle is to transfer the discs from the center peg to either of the other two pegs, ending with the discs arranged in the same order as at the start.



There are two rules:

1. Only 1 disc may be moved at a time.
2. A larger disc may never be placed on top of a smaller one.

Task 1: Take a few minutes trying to solve the puzzle. Record the number of moves for each attempt.

Task 2: Can you do the puzzle with 3 discs? Record the number of moves for each attempt. Is there a way to tell whether you have the minimum number of moves or not?

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Task 3: Try the puzzle using 2 discs, 4 discs and 5 discs. Is there a relationship between the number of discs and the minimum number of moves to transfer a pile? Collect the data and record on the table below:

# of disks [x]	1	2	3	4	5	
# of moves [y]						

Task 4: Record at least three different patterns from the table.

Task 5: Look for constant differences in the table. How does this table differ from previous data tables? What affect will this have on the function?

Task 6: If you started with 25 discs and it took 1 second for each move, estimate how long would it take to transfer the pile? [You will find the function rule later in the lesson; at that point you will be able to calculate the number of moves to determine amount of time].

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Why 3^0 is 1 (Marcia's way)

Part 1: Look at each problem. What relationship exists between the two exponents in the original problem and the exponent in the answer?

$$\frac{3^6}{3^2} = \frac{3 \times 3 \times 3 \times 3 \times 3 \times 3}{3 \times 3} = 3^4$$

$$\frac{3^6}{3^3} = \frac{3 \times 3 \times 3 \times 3 \times 3 \times 3}{3 \times 3 \times 3} = 3^3$$

$$\frac{3^6}{3^4} = \frac{3 \times 3 \times 3 \times 3 \times 3 \times 3}{3 \times 3 \times 3 \times 3} = 3^2$$

$$\frac{3^6}{3^5} = \frac{3 \times 3 \times 3 \times 3 \times 3 \times 3}{3 \times 3 \times 3 \times 3 \times 3} = 3^1$$

Write directions explaining a quick way to solve problems like these so you do not have to write out the middle step shown above. Apply your rule to these new problems

$$\frac{2^8}{2^5} = 2^{\square} \qquad \frac{5^{10}}{5^6} = 5^{\square} \qquad \frac{4^3}{4^2} = 4^{\square} \qquad \frac{9^6}{9^5} = 9^{\square}$$

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$$\frac{3^6}{3^6} = \frac{3 \times 3 \times 3 \times 3 \times 3 \times 3}{3 \times 3 \times 3 \times 3 \times 3 \times 3} = 1 = 3^{\square}$$

$$\frac{3^6}{3^7} = \frac{3 \times 3 \times 3 \times 3 \times 3 \times 3}{3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3} = \frac{1}{3} = 3^{\square}$$

$$\frac{3^6}{3^8} = \frac{3 \times 3 \times 3 \times 3 \times 3 \times 3}{3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3} = \frac{1}{3 \times 3} = 3^{\square}$$

$$\frac{3^6}{3^9} = \frac{3 \times 3 \times 3 \times 3 \times 3 \times 3}{3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3} = \frac{1}{3 \times 3 \times 3} = 3^{\square}$$

$$\frac{3^6}{3^{10}} = \frac{3 \times 3 \times 3 \times 3 \times 3 \times 3}{3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3} = \frac{1}{3 \times 3 \times 3 \times 3 \times 3} = 3^{\square}$$

$$\frac{2^5}{2^5} = 2^{\square} =$$

$$\frac{9^2}{9^2} = 9^{\square} =$$

I hope the patterns you observed here help you understand why 2^0, 9^0, 23^0 are all equal to one.

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